



Tribhuvan University

Institute of Science and Technology

A Final Year Project Proposal

On

“Crop Recommendation System”

Submitted to:

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1. Introduction

Agriculture is the basic source of food supply of all the countries of the world—whether underdeveloped, developing or even developed. The world population is estimated to be about 9.7 billion by 2025. This added with unpredictable weather conditions makes it difficult to ensure food sustainability. Fortunately, there is a solution for this problem as for many others. Crop Recommendation System takes the N-P-K (Nitrogen, Phosphorous and Potassium) and pH values along with the temperature, humidity values as input and recommends the optimal crop to the farmer, hence ensuring that the farmer takes an informed decision before cultivation. In this system, we train the model using Random Forest, Decision Tree, etc. We compare the accuracy of this models and choose the best out of it and store that specific model using pickle module and deploy the machine learning model using Flask [3] .

By exploring and evaluating the performance of these machine learning algorithms, agricultural stakeholders can gain insights into their suitability for crop recommendation. The comparative analysis of these algorithms will enable farmers, policymakers, and agricultural technology developers to make informed decisions regarding the adoption of specific algorithms based on their requirements and the available data. Ultimately, the application of machine learning in crop recommendation holds the potential to enhance agricultural practices, optimize resource utilization, and contribute to sustainable and efficient farming systems [5].

2. Problem Statement

Deciding on the crop to be cultivated is a difficult and onerous task for farmers, especially so for inexperienced ones. The knowledge that exists in agriculture is mostly based on heuristics or comes from anecdotal evidence. Such general knowledge does not take into account specific soil and climate conditions which are unique to every farm. Consequently, sub-optimal crop yield can be generated by this process and input might also be wasted.

A machine learning approach can provide a more accurate and personalized way of choosing crops as there are no such limitations. By examining important agricultural features like temperature, rainfall, pH of the soil, and nutrient content etc., the model will recommend crops suitable for any particular case basically putting its past experience into play. Using an

extensive dataset from Kaggle along with Random Forest algorithm the Crop Recommendation System offers data driven recommendations to aid farmers choose their best suited crops.

3. Objectives

- To recommend the crop to users based on the user input.
- To reduce the wrong choice on a crop by application of principles of precision agriculture.

4. Methodology

The system is designed to recommend crop to users based on the data (NPK of the soil, pH value, Humidity, Rainfall, Temperature) given by the user. The system uses Random Forest Classifier model for recommending the crop. Random Forest is a common machine learning algorithm that falls within the supervised learning category. It can be applied to both classification and regression issues in machine learning.

a. Requirement Identification

The expected requirements for this project are listed below:

Hardware Requirements:

- A server with enough processing power and storage.
- Devices with internet access.

Software Requirements:

- Programming Languages: Python backend (Flask).
- Database: MySQL is used to store user data and game information.
- Machine Learning Libraries for Algorithm Development

i. Literature review

The growing demand for sustainable agriculture practices and resource efficiency has led to a considerable increase in interest in the development of machine learning-based crop recommendation systems in recent years. In order to forecast the best crops based on variables like soil type, pH levels, and weather patterns, researchers have investigated a variety of machine-learning models, including as Decision Trees, Random Forests, and K-Nearest Neighbors.

In October 2020, Pradeepa Bandara, Thilini Weerasooriya, Ruchirawya T.H, W.J.M. Nanayakkara, Dimantha M.A.C, Pabasara M.G.P. proposed a system which used an Arduino board to collect data on several parameters like pH, moisture, humidity, temperature, erosion to predict suitable crop. The dataset contained around five different crops. The crop prediction was be done by Native Bayes Classifier, Support Vector Machine (SVM). By applying Native Bayes algorithm acquired higher productivity and accuracy of around 96% [1].

In April 2022, Lakshman Kumar Sreu and Sai Maanas Gandham have implemented the Decision tree and Random Forest classifier for the crop recommendation system [2]. In 2022 Ranjith Macharla, Dr. K Shirisha developed a Crop Recommendation System using Random Forest Classifier algorithm. They first compared the accuracy of other models like of Decision Tree Classifier in which they got the accuracy of 90%, KNN Classifier in which they got an average accuracy 97%, Random Forest Classifier in which they got an accuracy of 98.8%. 80% of dataset they used for the training the model and 20% of dataset they used for training the model. Among all the models, Random Forest Classifier got the highest accuracy so, they implemented their system using Random Forest Classifier model [3].

They used the dataset from the Kaggle and the Indian government websites. In March 2023, G. Buvaanyaa and Dr. S. Radhimeenakshi published research paper about Crop Recommendation System Using Random Forest Algorithm. Out of 100% samples, they trained 75% samples, and 25% samples used for the testing and found that the average accuracy of crop recommendation system was 91.99% [4].

In February 2024, researchers Monisha H M Dhanush Jogi K Adarsha Dalavai S M Manvith Dalli of B.M.S College of Bangalore, India published a research paper on A Web Based Crop Recommendation System Using Various Machine Learning Algorithms. They tested classifier algorithms like Logistic Regression, Decision Tree, Random Forest and XGBoost and found that among all the classifiers, highest accuracy was of the Random Forest Classifier which was 98.86%. Also, they concluded to use the Random Forest because overfitting will not happen easily and Random Forest gives the accurate results [5].

ii. Requirement Analysis

Functional Requirements

- User Authentication: Secure login for different users.
- Data Input: Users can input relevant data, including soil type, climate conditions.
- User Interface: A user-friendly dashboard to display recommendations.

Non-Functional Requirements

- Response Time: The system should make recommendations in a timely manner, particularly for users with limited internet connectivity.
- Data Privacy: The system must protect user data and follow applicable data privacy standards.
- Scalability: The system should be capable of handling increased data volumes and user loads while maintaining performance.

b. Feasibility study

A feasibility study is a process of analysis and assessment intended to ascertain the viability, practicality, and likelihood of success of a proposed project or proposal. It entails evaluating a number of variables, including operational, legal, technical, financial, and scheduling elements, to ascertain whether the project can be effectively finished within the allotted time, money, and resources.

i. Technical

The project is technically feasible; complies with current technology, including both the hardware and the software. The technical requirements for this project are mentioned below:

- A laptop with at least 8GB RAM, SSD (at least 256GB) and GPU.
- High speed internet

This application is compatible with most modern personal computers, with minimal hardware and software requirements.

ii. Operational

This project can be completed with minimal human resources and system needs. The project currently has three developers, which is sufficient for its needs. This project intends to develop a crop recommendation system based on the Random Forest Classifier Algorithm.

iii. Economic

The project is cost-effective as it utilizes open-source tools (e.g., Python, Jupyter Notebook) and free IDEs (e.g., VS Code) that can be downloaded for free. Many websites offer free website hosting until a storage limit is reached, making it an efficient option for this project. Also, the project's benefits readily outweigh the costs. As a result, the project is currently economically feasible.

iv. Schedule

Scheduling feasibility involves estimating the project's completion time. After all of these areas are considered, the feasibility analysis helps identify any obstacles the proposed project may face, including: Internal Project Constraints: Technical, Technology, Budget, and Resource.

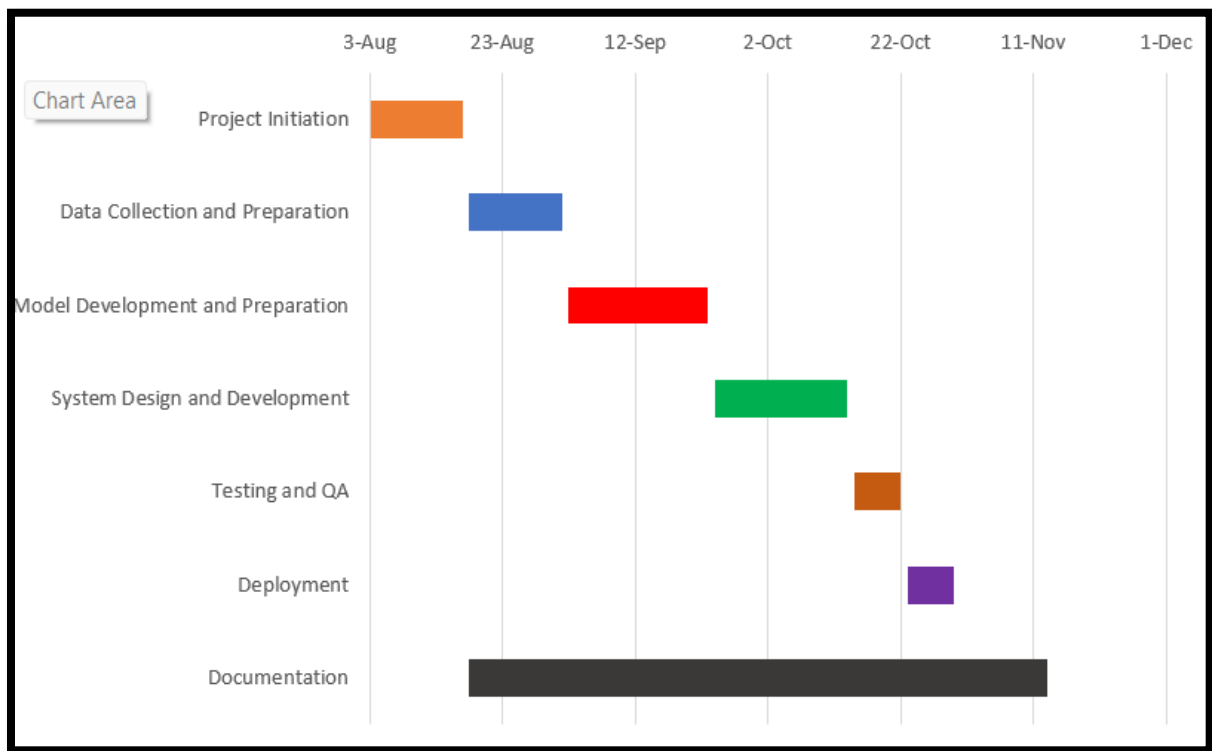


Figure 1 Gantt Chart

c. High Level Design of System

i. Methodology of the system:

The system is designed to recommend crop to users based on the data (NPK of the soil, pH value, Humidity, Rainfall, Temperature) given by the user. The methodology of the system is described below:

Data Collection

The information dataset is a yield proposal document that includes the dataset, which must be subjected to preprocessing. Datasets are used from the Kaggle.

Data Preprocessing

The input dataset undergoes various preprocessing procedures, including filling missing values, encoding straight-out information, and scaling values to the appropriate range.

Splitting Dataset into Training and Testing Dataset

The preprocessed dataset is then divided into two parts: preparation and testing, according to the preset split proportion. The suggested work uses a 75:25 split ratio, with 75% of the dataset used to prepare the troupe model and 25% used for testing.

Model Training and Testing

Random Forest Classifier is initialized and trained using the training dataset and then tested using the test dataset to find the accuracy, precision, etc. of the model.

Recommendations

Once trained and evaluated, the model can be used to make predictions about new data. This additional data may include soil parameters (such as pH and nitrogen content), climate variables (such as temperature and rainfall), and other important agricultural inputs.

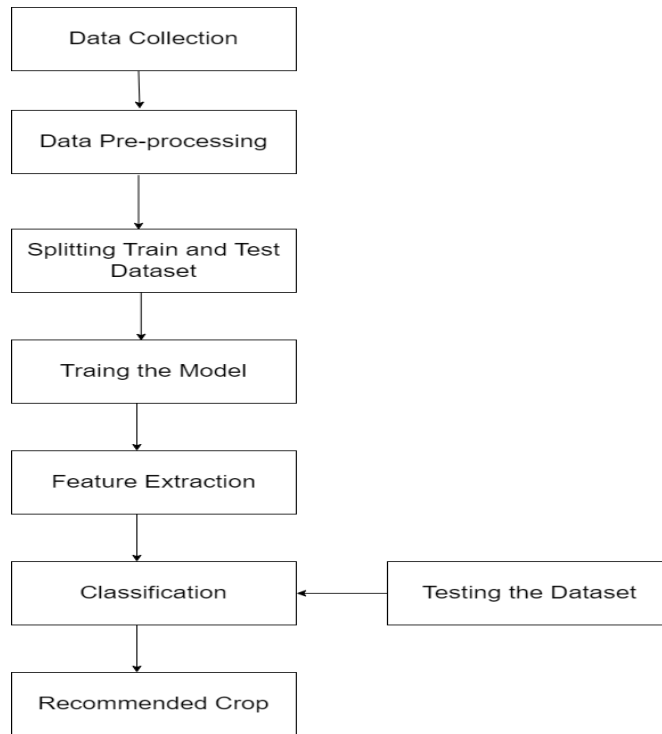


Figure 2 Proposed system flowchart

ii. Working Mechanism of Proposed System

The application is broken into two parts. The process involves two parts: developing the machine learning model and deploying it to production.

First part:

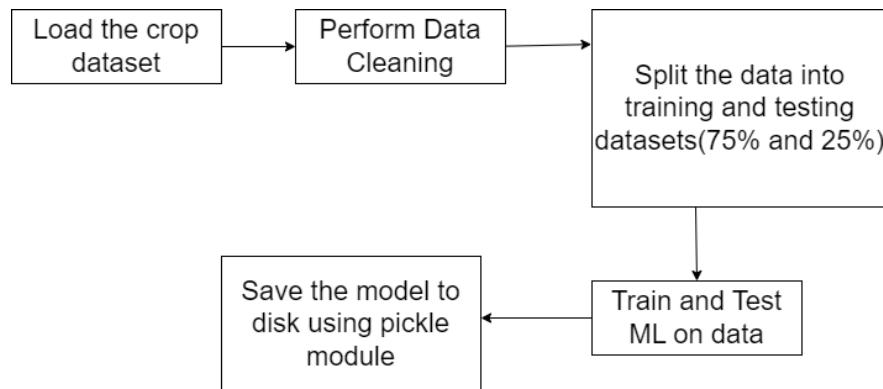


Figure 3 Crop Recommendation ML model Architecture

Second Part:

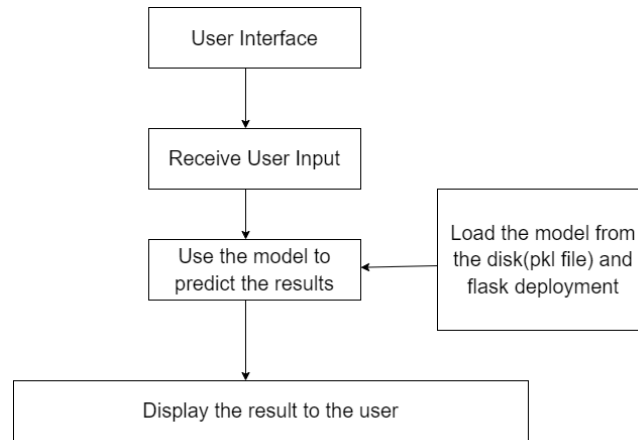


Figure 4 Web application Architecture

iii. Flowchart of the working of proposed web application system

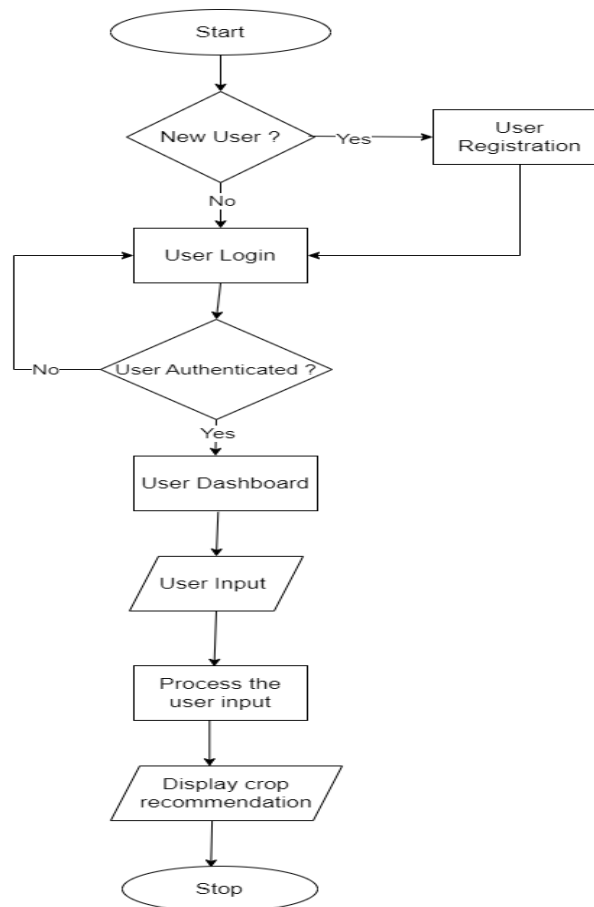


Figure 5 Flowchart of the proposed system

iv. Description of Algorithm

Random Forest is a common machine learning algorithm that falls within the supervised learning category. It can be applied to both classification and regression issues in machine learning. It is based on the concept of ensemble learning, which involves merging numerous classifiers to solve a complex problem and improve model performance. According to its description, "Random Forest is a classifier that contains a number of decision trees on various subsets of the given dataset and takes the average to improve the predictive accuracy of that dataset." Rather than depending on a single decision tree, the random forest collects forecasts from each tree and predicts the ultimate output based on the majority of predictions [6].

The more trees in the forest, the higher the accuracy and the lower the risk of overfitting. The following diagram demonstrates how the Random Forest algorithm works:

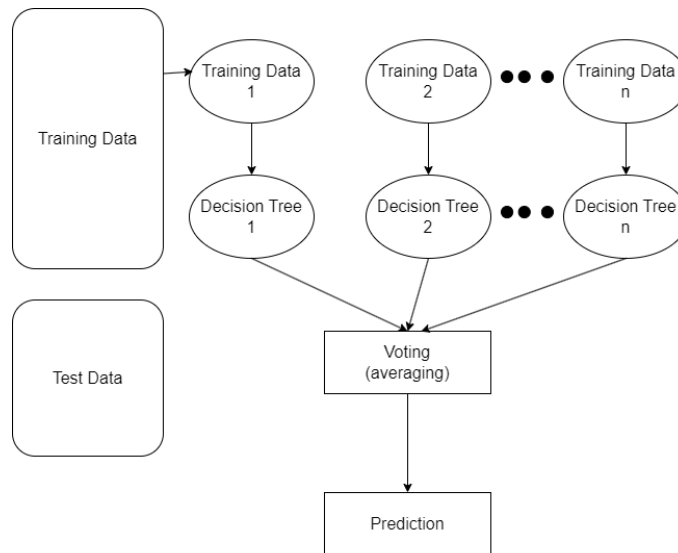


Figure 6 Working of Random Forest Algorithm

Algorithm for Random Forest in Machine Learning:

Step 1: Select random samples from a given data or training set.

Step 2: This algorithm will construct a decision tree for every training data.

Step 3: Voting will take place by averaging the decision tree.

Step 4: Finally, select the most voted prediction result as the final prediction result.

5. Expected Outcome

- The system will be able to give accurate and relevant crop recommendation to the user based on the environmental conditions.
- The system will help in promoting the environmentally friendly crop choices.

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